



Towards a neuroadaptive augmented reality piano tutor: opportunities and challenges

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Abstract

The fusion of mixed reality with EEG-based neuro-adaptive systems are becoming increasingly popular in the field of learning and education. Their ability to detect changes in brain activity in real-time makes them unique and enables monitoring different mental states like mental workload (MWL) and mental fatigue (MF) aiming to increase learning performance. We present a neuro-adaptive system that combines an augmented reality (AR) piano tutorial with online EEG measurements of its user, delivered by a passive Brain Computer Interface (BCI) system. The MWL was measured by means of EEG and a Filter Bank Common Spatial Patterns algorithm (FBCSP) was trained to differentiate between low and high levels of MWL. Low levels were connected to the 0-back task and high levels to the 2-back task. The n-back task was a calibration task to train a binary Machine Learning (ML) classifier to differentiate between low and high MWL. This trained ML algorithm was then used as the central element of the passive BCI, which constantly classified small temporal windows of the EEG during the piano tutorial and adapted the difficulty of the tutorial. 22 Participants were randomly separated into two groups: adaptive and non-adaptive piano tutorial. The results of the non-adaptive group showed significantly higher levels of classified MWL throughout the piano tutorial. Additionally, a band power (BP) analysis was conducted, which revealed increased levels of MWL and MF towards the end of the piano tutorial for both groups. For the non-adaptive group, signs of MF were detected in all levels except for level 2 and 6. This inferred that the adaptive group experienced less MF over the tutorial than the non-adaptive group, suggesting that the adaptive difficulty helped reduce MF. Although no significant differences were observed between the groups in terms of learning performance but participants in the adaptive group reported higher satisfaction and engagement during their piano play. Our findings suggest the potential of passive BCIs in enhancing learning experiences and paving the way for future studies combining mixed reality applications with passive BCI technology.

Keywords EEG · Augmented reality · Mental workload · Mental fatigue · Passive BCI

1 Introduction

Neuro-adaptive Systems are a growing field that leverages advancements in Brain-Computer Interfaces (BCIs) and physiological monitoring to enhance personalized and adaptive learning.

experiences. By dynamically adapting the difficulty of tasks based on real-time assessments of cognitive states such as Mental Workload (MWL), fatigue, and motivation, Neuro-adaptive systems aim to optimize learning conditions and reduce cognitive overload. This adaptive approach holds the potential to revolutionize traditional educational and training methods by offering tailored experiences that address the unique needs of each individual (Armani et al. 2023). Such systems generally use the methodology of brain-computer interfaces (BCI) (Millán et al. 2010; Müller-Putz et al. 2022; Wolpaw et al. 2002). BCIs can be implemented as active BCIs, which are actively controlled by the user (e.g. motor imagery), as reactive BCIs, controlled by the reaction of the user to external stimuli (e.g. Event-Related Potentials) or as passive BCIs (pBCI)

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(Zander and Kothe 2011) based on the cognitive feedback of the users' brain activity (e.g. mental state), as well as other physiological measurements (Kawala-Sterniuk et al. 2021; Müller-Putz et al. 2011, 2015; Pfurtscheller 2010). There are also other approaches, not relying on brain activation but using physiological signals exclusively. Some current adaptive approaches just rely on physiological signals, like galvanic skin response, heart rate or heart rate variability. Of course, such signals are easy to interpret and directly observable. They also provide an indication of the actual mental state of the user but they lack in specificity related to the underlying cognitive and emotional states (Hettinger et al. 2003). Additionally, the timing associated with changes in biological signals may render them not suitable in situations in which rapid adaptive responses are required. Therefore, we use brain activity measured with EEG to develop our neuro-adaptive system. It has the potential to close key gaps in current adaptive approaches by addressing challenges related to real-time adaptation, user variability, and system robustness.

Specifically, pBCIs are the core part of the current work since they derive their outputs from neuro-physiological activity with implicit information on the actual state of the user, enabling applications spanning a variety of domains, such as workload assessment (Wriessnegger et al. 2021; Andreessen et al. 2021), gaming (Arico et al. 2018), and marketing (Aldayel et al. 2021; Wriessnegger et al. 2015, 2017). A potential, high-value use of a pBCI is the monitoring of mental states, like fatigue or mental workload, in high-risk surroundings, as for air traffic controllers, pilots, and other industrial operators. For those occupations, human error could be potentially fatal, so pBCIs could detect changes in the mental workload or fatigue level of persons and initiate measures to prevent dangerous events and following improve human-computer interactions (Bagheri and Power 2022). BCIs additionally offer tailored learning experiences in education by adjusting educational content based on real-time brain activity analysis (Beauchemin et al. 2024; Walter et al., 2017; Petit et al., 2026

1.1 Mental states

Particularly two mental states affecting people during a cognitively demanding task are mental workload (MWL) and mental fatigue (MF). Even though in the past years there is growing interest in defining MWL, there is no clearly defined, universally accepted definition available (Hancock et al. 2021). Quantifying mental workload is relatively straightforward, but operationalizing it presents significant challenges. The lack of a cohesive framework for understanding human mental workload—its causes, processes, and measurements—has led to a concept that is abstract,

loosely defined, and backed by an expanding yet inconsistent body of findings (Van Acker et al. 2018). Complicating matters further is the fact that mental workload, like stress and fatigue (Matthews 2002), is a transactional construct, representing the interaction between an individual's capacities and the demands of a specific task. Moreover, MWL describes the collective, multidimensional demands that an individual must meet to complete a task in alignment with their skill level (Hancock and Chignell 1986; Paas et al. 2003). These external demands vary in their physical, mental, temporal, and emotional challenges, among others. To meet these demands and accomplish the task effectively, individuals need a sufficient level of skill, often developed through experience and learning. However, task performance can be achieved by people with varying skill levels; a less skilled individual might exert more effort to succeed, while a highly skilled person may reach the same outcome with less effort. Cognitive workload, then, reflects the effort required to manage task demands, influenced by factors such as task complexity, skill, and experience. Following that MWL can be defined as a measure of task difficulty. Alternatively, it can be described as the load in working memory, the number of tasks to be performed simultaneously or as a measure of mental resources engaged in a task. It has been documented that MWL can negatively influence task performance (Xie and Salvendy 2000). For example, when MWL is high, a higher level of confusions can be reached by individuals with a consequent decrement in their performance while processing information resulting in increased probability of making mistakes.

It is undisputable that MWL is a multidimensional construct influencing several disciplines from human factors and ergonomics to neuroscience and neuro-ergonomics. There are a lot of uncertainties in its definition and operationalization. In a previous review article Longo et al. (2022) propose the following novel inclusive definition of MWL: *"Mental workload (MWL) represents the degree of activation of a finite pool of resources, limited in capacity, while cognitively processing a primary task over time, mediated by external stochastic environmental and situational factors, as well as affected by definite internal characteristics of a human operator, for coping with static task demands, by devoted effort and attention"*. This novel definition explicitly mentions the multidimensionality of the construct which is influenced by several factors, including the environment, the capacity of the human cognitive system and the related resources. It specifically delivers an operational characterisation of MWL in the context of learning by providing several influencing subcomponents, like situational factors, individual characteristics like neuro-physiological factors. They claimed that these measures, specifically EEG and pBCIs, can support an accurate model of MWL together

with self-reports. The performance and safety of users of technological devices in their everyday life can be compromised by mental overload as a result of high MWL.

On the other hand, MF or reduced alertness arises from increasing time-on-task (TOT). Subjectively, it can be described as the lack of energy or the feeling of weariness during long periods of cognitive activity. In a more objective view, it is connected to exhaustion and tiredness leading to a decrease in task performance and commitment. High levels of MF may hinder the user to complete a task requiring self-motivation (Roy et al. 2013; Wriessnegger et al. 2021). From a neurocognitive standpoint, mental fatigue during learning is associated with reduced cortical efficiency, increased reliance on compensatory control mechanisms and as a shift from task-relevant processing to resource conservation (van der Linden 2011). MF is also described as a decline in mental performance caused by prolonged cognitive activity (Boksem and Tops 2008) or viewed as a temporary depletion of cognitive resources required for effective learning (Ackerman 2011). In Neuroergonomics it is defined as a state of reduced neural efficiency affecting attention and executive control systems (Warm et al. 2008; Mehta and Parasuraman 2013).

1.2 Measures of Mental Workload (MWL) and Mental Fatigue (MF)

Different methods exist to measure the human MWL and MF, which can be categorised into three main groups. Subjective measures, usually in the form of a post-task survey, make use of the subjective feedback provided by humans interacting with an underlying task. Some well-known surveys for MWL include the NASA-TLX (Hart 1988), the Workload Profile (WP; Tsang and Velazquez 1996) and Subjective Workload Assessment Technique (SWAT; Reid 1988). For MF, the Visual Analog Scale to Evaluate Fatigue Severity (VAS-F; Lee et al. 1991) questionnaire is an option. Task performance measures use the objective measurement of a human's performance in an underlying task. Some examples include reaction times (RTs), completion time of a task, number of errors and tapping errors. Physiological measures stem from the analysis of physiological responses of the human body, for example EEG, MEG, brain metabolism, pupil diameter, heart rate variability, et cetera (Raufi and Longo 2022; Wriessnegger et al. 2021).

Measuring and assessing learners' MWL is crucial for optimizing learning environments and improving their educational experiences and outcomes. As already mentioned above self-reported measures are the most commonly used method for evaluating MWL in the development of various educational technology tools, as they provide insights from the learners' perspectives (Anmarkrud et al. 2019; Brunken

et al. 2003; Mutlu-Bayraktar et al. 2019). However, these measures cannot objectively and accurately quantify the mental effort exerted during the learning process. Self-assessments depend on the learners' subjective awareness and require a deeper reflection on their learning experiences. It is undisputable that subjective measures provide valuable insights into the perception of cognitive load, but they fail to fully represent the learners' dynamic cognitive states which are essential for adapting teaching methods. Employing neuro-physiological measurement tools, such as eye tracking data, hormone levels, heart rate and heart rate variability, galvanic skin response (GSR) and brain activity (Riedl and Léger 2016; Brunken et al. 2003), can offer objective, more reliable assessments of MWL and complementing traditional methods.

Numerous studies show that band power (BP) features extracted from EEG can successfully detect mental states (Hassan et al. 2024). Borghini et al. (2014) did a meta-analysis of papers related to MWL, MF and situational awareness measured with EEG, EOG and the heart rate. They concluded that high mental workload correlates to an increase of theta BP (4–8 Hz) and a simultaneous decrease of alpha BP (8–12 Hz). Furthermore, the transition between MWL and MF is characterised by an increase in the theta as well as delta (1–4 Hz) and alpha BP. Similar findings were reported by Babiloni (2019). It was concluded, that higher MWL is associated with increased BP in the theta band in the frontal cortical area and a decreased BP in the alpha band in the parietal cortical region. Klimesch et al. (1999) reported an increase in MF to be associated with a BP decrease in the frequency bands above 12 Hz and a simultaneous increase in the BP of the frequency bands lower than 12 Hz.

1.3 Decoding Mental Workload (MWL) and Mental Fatigue (MF)

A common way detecting mental states in EEG data is using machine learning (ML) algorithms (Acı et al. 2019; Müller et al. 2008; Rahman et al. 2022; Appriou et al. 2018; Aksu et al. 2024; Ke et al. 2023). For example, Safari et al. (2024) propose some methods to assess MWL from EEG data which use effective brain connectivity for feature extraction and a hierarchical feature selection algorithm to select the most significant features to be finally classified by various machine learning models. Their results revealed that Support Vector Machine (SVM) work better than others and could classify the mental workload levels with accuracy equal to 89.53% (± 1.36). Others (Mohanavelu et al. 2022) used classification algorithms, like LDA, SVM, and k-nearest Neighbor (KNN), to classify the pilots' cognitive workload. The results demonstrated that LDA and SVM, with an accuracy of 75%, were more consistent classifiers

compared to the k-NN classifier, which achieved an accuracy of 60%. Similar ML approaches are used to decode MF in EEG. In a review study by Hooda et al. (2022) different ML algorithms are compared to find the most promising approach for MF detection. Specifically for EEG signals the Gradient Boosting Decision Tree (95%) and the Linear Regression Model (90%) works best. Most of the studies are aiming to find a reliable method to detect driver's fatigue to prevent car accidents (Fouad 2023; Hasan et al. 2023). Hussein et al., reported in their review paper about thirteen articles used SVM to classify EEG data detecting drowsiness state and achieved different classification performance ranging from 72 to 97%. Reasons for this inconsistency are an insufficient number of subjects leading to unreliable precision or lacks in the EEG channel selection. By comparing different methods and features that can be used for fatigue detection, they came to the conclusion that there is no single feature that can reliably detect fatigue. The methods that are currently used for the detection of fatigue rely mostly on a single parameter. But fatigue is a more complex phenomenon, it cannot be detected accurately with a single feature. They suggested methods which can use a combination of multiple features which are more credible. Specifically for real-time implementation of fatigue detection systems this would improve reliability and robustness. Following that the current study focused on MWL detection for the real-time implementation of the piano tutor. EEG features reflecting MF will be analysed and presented only as offline results.

A shrinkage LDA (sLDA) classifier was used to unravel the MWL in participants (Katmah et al. 2021; Sardar et al. 2022) during playing piano. Following the suggestion by Gerjets et al. (2014) a cross-task classification, which involved a training task that induced MWL in well-defined differences and a subsequent different target task, where the MWL was analyzed. They theorized, that both tasks should be as closely related as possible in terms of MWL they induce, and that the classifier should be avoided of picking up task-specific features. In this study, cross-task classification was used, where the training and target tasks were implemented as an n-back task and a piano tutorial, respectively.

1.4 Augmented reality (AR)

Virtual Reality (VR) and Augmented reality (AR) applications have recently started exploring adaptive training systems based on EEG measures (Dey et al. 2019) and mental workload assessment (Chiossi and Ou 2023; Luong et al. 2020). While many VR applications can benefit from using EEG measures (Choi et al. 2023), we pay particular attention to AR training for learning music instructions. AR has been shown to replace or enhance more traditional learning

experiences when used in higher education scenarios (Liarokapis 2010). Specifically, for musical self-teaching, AR visualizations can help beginners overcome the initial difficulty of translating written notation, most commonly sheet music, to physical keys on a keyboard by pairing instructions with a user's view of the instrument (Chow et al. 2013). While this work is focused on novice users, music tutorials in general address a large community and enable a highly variable level of complexity. Whereas the potential of pBCIs in the learning context has already been recognized by some studies (Yuksel et al. 2016; Walter et al. 2017; Kosmyna and Maes 2019; Beauchemin et al. 2024; Apicella et al. 2022), their use in the educational context is still limited and inconsistent. Other studies focusing on neurofeedback (NF) training have shown that AR is less nausea-inducing and could have positive effects on NF performance due to its interesting and immersive character. The usability of AR in EEG-NF trainings has been shown by Berger et al. 2024; reporting less nausea symptoms compared to VR and a positive influence on NF training performance (Berger et al. 2024; Gorman and Gustafsson 2022).

The advantages to combine a pBCI with AR technology in education is manifold: (1) It provides an enhanced user interaction and immersion by just overlaying digital information in real-world environments. Specifically, it enhances engagement by making the learning process more interactive and enjoyable. (2) By using the mental states of users as additional factor more personalized learning or training can be created and the difficulty level of the song to play can be dynamically adjusted. The advantage is that it adapts to the learner's pace, offering customized lessons based on skill level and progress. By modifying the difficulty levels dynamically, it is possible to keep students challenged but not overwhelmed. (3) By combining both technologies, the learning process could be optimized regarding MWL information, UI complexity and also attention can be improved by reduced distraction. (4) Overall AR-guided BCI training can improve mental focus, memory, or motor skills by providing real-time feedback based on brain activity. (5) It delivers a faster and independent interaction because direct brain control removes in combination with the piano tutorial excludes the need for traditional methods (teacher or piano classes), speeding up the learning process since it can be used at any time without restriction to opening hours.

To the best of our knowledge, our study represents for the first time the fusion of BCI technology and AR using real-time data to classify MWL aiming to enhance learning performance during the piano play.

A preliminary version of the proposed neuro-adaptive system was previously presented as a conference poster in Maitz et al. (2023). In the present paper, we extend this work by reporting the complete experimental study and

providing a detailed description of the system, methodology, and results. In addition to the online mental workload (MWL) classification, we present comprehensive neurophysiological analyses of mental workload and mental fatigue (MF), behavioural performance measures, and subjective evaluations using the NASA-TLX and SUS questionnaires, enabling a more thorough assessment of the proposed neuro-adaptive AR piano tutoring system. Following that we are aiming to answer the following research questions: Is it possible to successfully combine an EEG-based pBCI system in online mode with the AR application to learn playing piano. Does adapting the difficulty of the piano tutorial to the learner's real-time MWL enhance the learning performance? Accordingly, we formulated the following hypotheses: Persons using the neuro-adaptive method will show significantly better piano performance compared to persons assigned to the non-adaptive group. Furthermore, we hypothesize that MWL will be significantly higher in the non-adaptive group compared to the adaptive group which is associated with a band power decrease in the alpha frequency band (8–12 Hz) and an increase in the theta frequency band (4–8 Hz). Regarding the MF detection we assume an increase in the BP frequency bands below 12 Hz and a simultaneous BP decrease in the frequency band higher than 12 Hz which is associated with MF especially for the non-adaptive group. Both, the reduced MWL and MF BP changes will support the effectiveness of a neuro-adaptive system in learning environments.

2 Methods

2.1 Participants

Twenty-nine healthy participants, mainly students of the Graz University of Technology, participated in this study. Seven participants had to be excluded due to excessive artefacts in their EEG data. Specifically, in two of the seven rejected measurements, the machine learning classifier could not be trained with data from the n-back task because the automatic bad EEG epoch rejection algorithm eliminated too much information. In the other five cases, the EEG data were dismissed based on visual inspection after the measurement due to artefacts in the signal. Consequently, the data from the remaining 22 participants (8 females, 14 males; ages ranging from 19 to 50 years; with a mean age of 28.22 years and a standard deviation of 6.13 years) were considered for the analysis in this study. Before the measurements began, participants were asked to complete a demographics survey that included information on gender, age, profession, use of glasses or contact lenses, handedness, piano skills, and experience and usage of Mixed Reality

(MR) applications. Participants rated their piano skills, MR experience and usage on a scale from one to five, where one indicated “very low” and five “very high.” Twenty-one out of 22 participants rated their piano skills at the lowest level (1), with only one person rating it a 2. The mean score for MR experience was 1.68, with a standard deviation of 1.14, and for MR usage, it was 1.23, with a standard deviation of 0.42. Figure 1 shows the setup of the system exemplarily on one participant.

All participants were fully informed about the experimental procedures in advance and voluntarily provided written informed consent. They were also informed about the susceptibility of EEG recordings to various sources of noise, which was demonstrated using live EEG recordings while participants performed different eye movements and clenched their teeth. Participants were instructed to avoid these actions and to remain as still and relaxed as possible, minimizing head and facial movements. Participants were randomly assigned to one of two groups: adaptive (based on MWL during tutorial) or non-adaptive (only tutorial). Therefore, all n participants were assigned a random number from 1 to n , even numbered participants were assigned the adaptive group, odd numbered participants the non-adaptive group. The randomisation process was done in Excel with the RAND() function.

The study was approved by the ethics committee of Graz University of Technology (Votum: EK-27/2024) and was in accordance with the ethical standards of the Declaration of Helsinki.

2.2 Experimental design

Each participant was sitting in front of a 45 cm high table, with a Yamaha P-45 B piano (Yamaha Corporation, Hamamatsu, Japan) on top of it. The Meta Quest 2 HMD (Reality Labs, United States) was the chosen modality to create an AR environment. The EEG was recorded with 34 active electrodes, which were mounted according to the 10–20 international system and connected to a wireless amplifier (LiveAmp: Brain Products GmbH, Gilching, Germany). More details regarding signal recording can be found in Sect. 2.5. Figure 2 gives an overview of all components and their connection of the developed neuro-adaptive system.

This study was split into two different tasks: first, participants conducted the n-back task (Kirchner 1958) and then the piano tutorial, which was realised as a passive BCI and consisted of ten levels. During both tasks the subjects were wearing the EEG cap, and information was delivered through the AR headset. Feedback from the participants was acquired through pressing keys on the piano. The experimental procedure is illustrated in Fig. 3. First, participants familiarized themselves with the n-back task by completing

Fig. 1 The experimental setup, with the participant seated in front of the piano wearing the electrode cap and the Meta Quest 2



Fig. 2 General overview of the system pipeline illustrating the main components and the data flow of the experiment



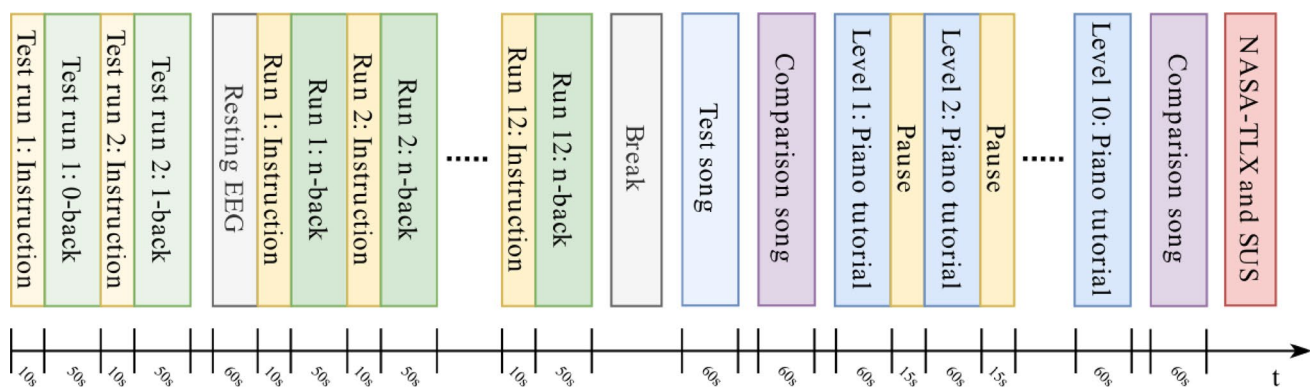


Fig. 3 Timing of the experimental paradigm

a test run consisting of a 0-back and a 1-back task to ensure they understood the procedure. Following this, the actual n-back trials began. These trials included a one-minute resting EEG session where participants were asked to focus on the wall and let their minds wander. This was followed by 12 randomized runs of 0-, 1-, and 2-back tasks (Fig. 5). Upon completion of the paradigm, the script automatically initiated training for the machine learning classifier, realized as a sophisticated Filter Bank Common Spatial Patterns algorithm (FBCSP), using only the EEG data recorded during the session. During this brief period (approximately 30 s), participants were offered a break, allowing them to remove the Meta Quest 2, have a glass of water, and enjoy some sweets. After the FBCSP algorithm finished training, the next step involved the test song, similar in purpose to the initial n-back test run. This step was designed to help participants become comfortable with the piano app and its controls. Before and after a brief piano tutorial, participants were instructed to play the “comparison song.” This song was played both before and after the main piano tutorial to track participants’ progress in piano learning. Following the first comparison song, participants began the main tutorial, which consisted of 10 levels.

In the non-adaptive group, each tutorial level introduced a new song of increasing difficulty (i.e., the song number matched the level number), with song 1 being the easiest and song 10 the most challenging. In the adaptive group, the song presented at each level was determined by the participant’s mental workload (MWL), which was continuously monitored using the trained FBCSP algorithm. The average MWL during a given level was used to determine the difficulty of the subsequent song. If MWL was high, the same song was repeated; if MWL was low, the next, more challenging song was introduced. As a result of this adaptive design, the song number and level number did not necessarily correspond. To prevent excessive repetition due to consistently high MWL, a more difficult song was introduced after three consecutive levels with high MWL. After

completing all 10 tutorial levels and the second comparison song, participants removed the VR/AR glasses and EEG cap and completed the NASA Task Load Index (NASA-TLX; Hart 1988), which assesses perceived mental workload, and the System Usability Scale (SUS; Brooke 1996) to evaluate the tutorial.

2.3 AR application

The AR piano teaching application was implemented in C# and Unity and for visualization we selected the Meta Quest 2 headset. This device enables video see-through AR via its Passthrough API functionality, which blends virtual and real content and displays the result on its screens. The cameras on the headset produce a grayscale image of the environment with the virtual augmentations being in colour (Fig. 4).

The virtual content is seamlessly integrated into the real scene to align with the actual image. The positioning of the real piano in relation to the headset is determined by placing the device’s controller at a specific location on the instrument, and its movement is tracked through the headset’s spatial tracking system. In augmented reality (AR), piano instructions are presented through small rectangular overlays that mirror the keys of the piano, each corresponding to individual notes. These overlays, depicted in Fig. 4, are color-coded based on whether they represent a white or black key and move gradually towards the physical keyboard. Users are prompted to play a piano key when these rectangles align with the corresponding real key. The holographic keys (brighter and darker green) moved towards the player and once they hit the horizontal green line, the player was supposed to press the corresponding piano key. The turquoise holographic rectangle is the feedback of the App of the user pressing a button on the piano. This visual representation enables users to anticipate upcoming notes before playing them. The rhythm video game “Keyboardmania” (Konami, Tokyo, Japan) and the training program “Synthesia” (Synthesia, London, United Kingdom) serve as

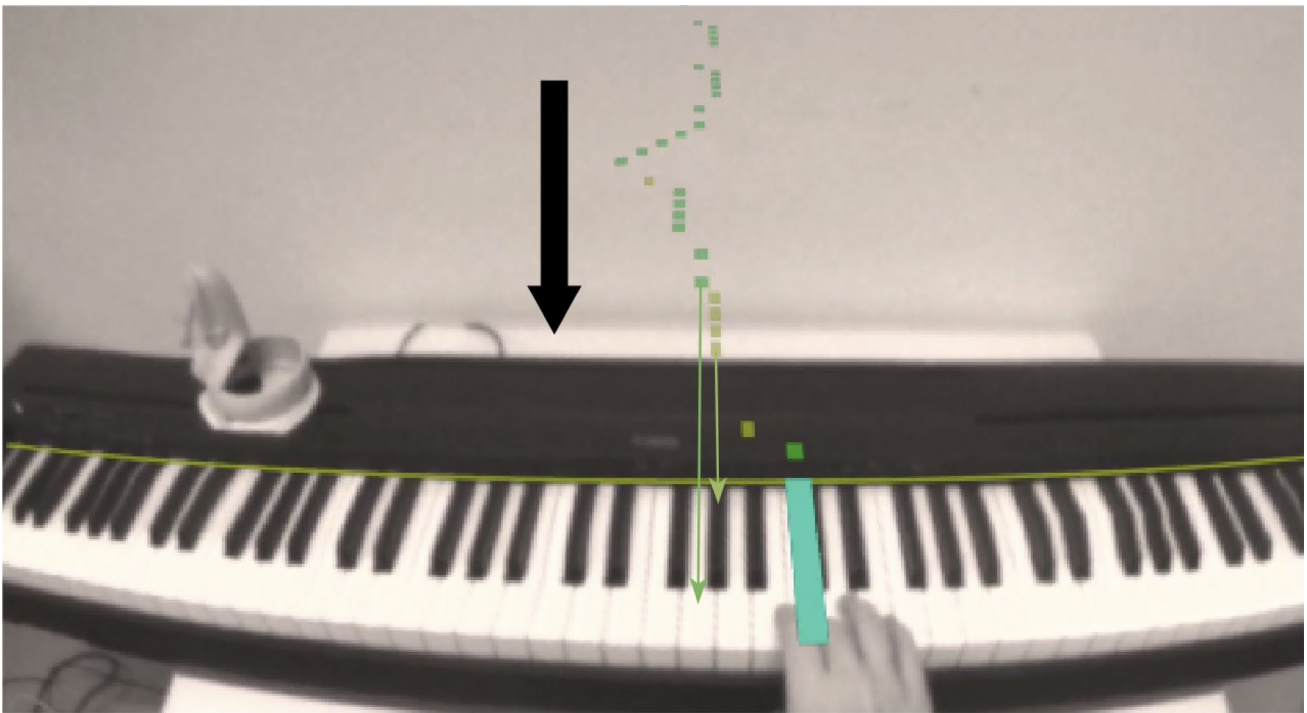


Fig. 4 Screenshot illustrating the AR application of the piano tutorial

examples of desktop software that utilize a similar style of visualization. Our piano establishes a connection with an attached laptop computer through USB-to-Host and transmits signals to the Meta Quest 2 headset via wireless UDP whenever a key is pressed. This functionality allows the teaching application to offer feedback to the user by displaying a coloured marker over the played key, as illustrated in Fig. 4, and recording previously played notes. In our implementation, lesson content is generated by parsing MIDI files. These files contain information about the notes to be played at any given time, which our tool utilizes to create the aforementioned colourful markers and position them within the scene accordingly. The ease of generating additional lesson content and the efficiency of memory usage enable users not only to share lessons but also to benefit from existing learning material available online (Skreinig et al. 2023). Websites like PianoMidi (<http://www.piano-midi.de/>) or BitMidi (<https://bitmidi.com/>) provide songs in MIDI format, storing notes and their timing. Typically, these files require no modification to be used by our system, except in cases where they contain multiple tracks for other instruments. In such instances, piano tracks can be exported to a separate file using online tools like Online Sequencer.

2.3.1 The n-back task

Participants performed the n-back task in varying difficulties before doing the piano tutorial to train the classifier.

Different n-back conditions were used to mentally challenge participants and induce low (0-back), intermediate (1-back) and high (2-back) levels of MWL (Fig. 5). The goal of the task was to decide, whether the currently shown letter was a target or non-target letter.

Participants pushed a random key at the right (target) or left (non-target) half of the keyboard, to confirm their decision. The last key, which was still a target key was the C4 key. A letter was considered a target letter during the 1- and 2-back condition, if the currently shown letter was the same as the letter n-letters back. If this was not the case, it was considered a non-target letter. In case of the 0-back condition, the target letter was simply the letter “X”, and every other letter was considered a non-target letter. For all conditions, a total of eight different consonants were used. Vowels were excluded to reduce the chance of letters randomly creating words and participants creating chunking strategies which would reduce.

mental effort (Brouwer et al. 2012). For 0-back, those letters were: “B”, “F”, “H”, “J”, “M”, “Q”, “R” and the target letter “X”. For the 1- and 2-back task, the used letters were the same as in the 0-back case, except the letter “X” was swapped with “T”, to avoid possible confusions with the target from the 0-back. For each run, 25% of letters were targets. Figure 5 illustrates an example of the different conditions where a green check mark means that the letter is a target letter, and a red X means that it is a non-target letter.

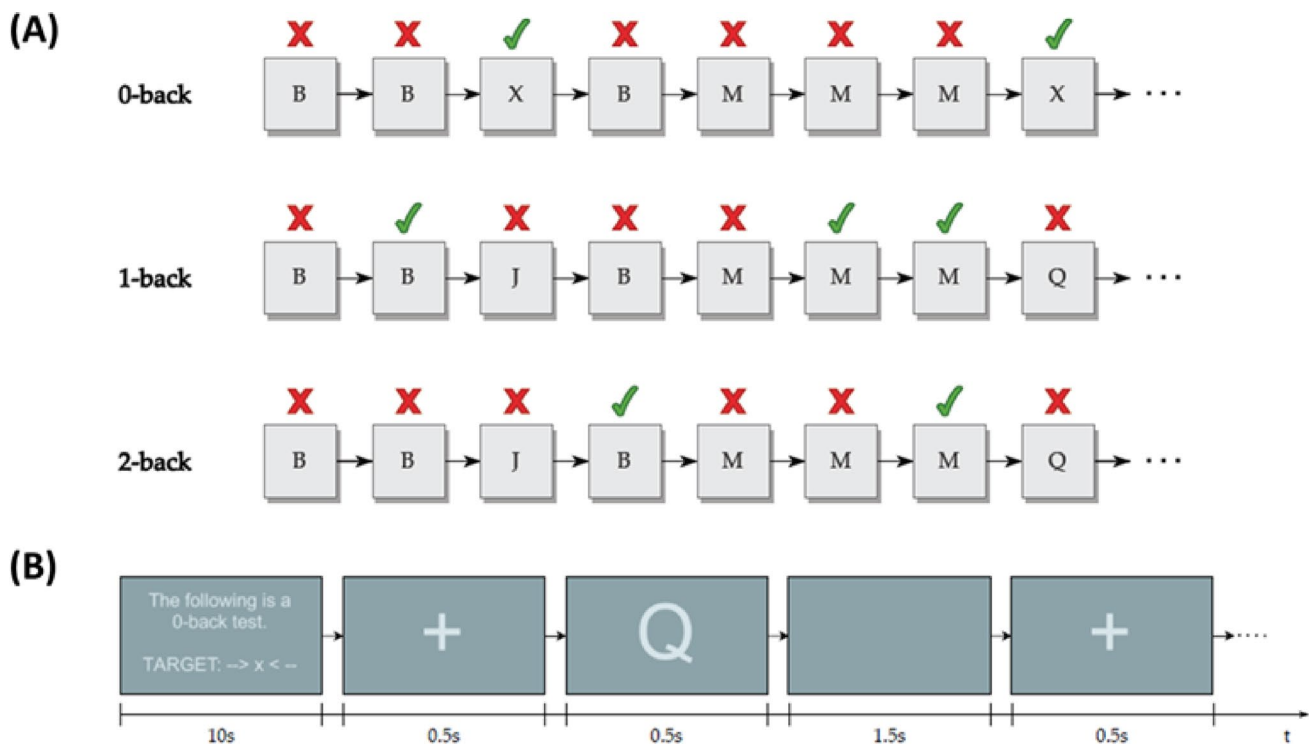


Fig. 5 **A** Example of the different n-back conditions. **B** One run of the n-back paradigm including the timing as seen from the participants through the Meta Quest 2

2.3.2 Piano tutorial

After all n-back runs and a brief break, participants started with the piano tutorial section of the experiment. The tutorial was divided into different songs, each lasting 60 s. To prepare the user for the upcoming task, each song started after a 3 s countdown. Subsequently, small holographic keys began to move towards the real keys on the piano, approaching the player (see Fig. 4). These keys took the form of small rectangles, bright green and elongated for black keys, and dark green and square-shaped for white keys. Upon reaching the green horizontal line, representing the top border of the piano keys, the corresponding physical key was overlaid with a hologram, signalling the participant to press the corresponding button. Once a button was pressed, the tutorial provided feedback by overlaying the pressed button in turquoise (Fig. 4). Each holographic key represented a tone of the song. Throughout every segment of the piano tutorial section, including the test song, comparison song, and the 10-level tutorial, users were guided by the aforementioned holographic aid. In total, there were twelve different songs, comprising one test song, one comparison song, and ten piano tutorial songs, selected based on their subjective difficulty. Both the adaptive and non-adaptive groups played a total of ten levels consecutively (excluding the comparison song and test song). Each level lasted 60 s and consisted of a

specific song. A 15-second pause interval between the songs allowed users time to relax.

The previously mentioned three seconds countdown was included in the 15 s pause frame. In the adaptive group, in case the MWL was high, participants had to play the same song again the next level and in case the MWL was low, the next harder song would be played.

In the non-adaptive group, the level index always coincided with the song index, meaning that the tutorial was getting constantly harder. The main part of the tutorial lasted exactly 12 min and 15 s. When participants felt ready to start, the second run of the comparison song was started, which was the last part of the piano tutorial part of the experiment.

In the following, the names of the tutorial songs are stated, with the download link of their MIDI file cited. Most songs were edited with the free software Online Sequencer (<https://onlinesequencer.com>) to fit the tutorial in a better way. The test song, which was the easiest song to play and got the participants used to the piano tutorial, was “Alle meine Entchen” (<https://www.lieder-archiv.de>). The comparison songs were both the same nursery rhyme called “Row, Row, Row Your Boat” (<https://bitmidi.com>) and was selected to be in an easy to intermediate difficulty, around the difficulty of song 2 and 3 in the piano tutorial. The songs of the piano tutorial from song 1 to 10 in rising difficulty were: “Twinkle, Twinkle, Little Star”, “Flohwalzer”, “Old MacDonald Had

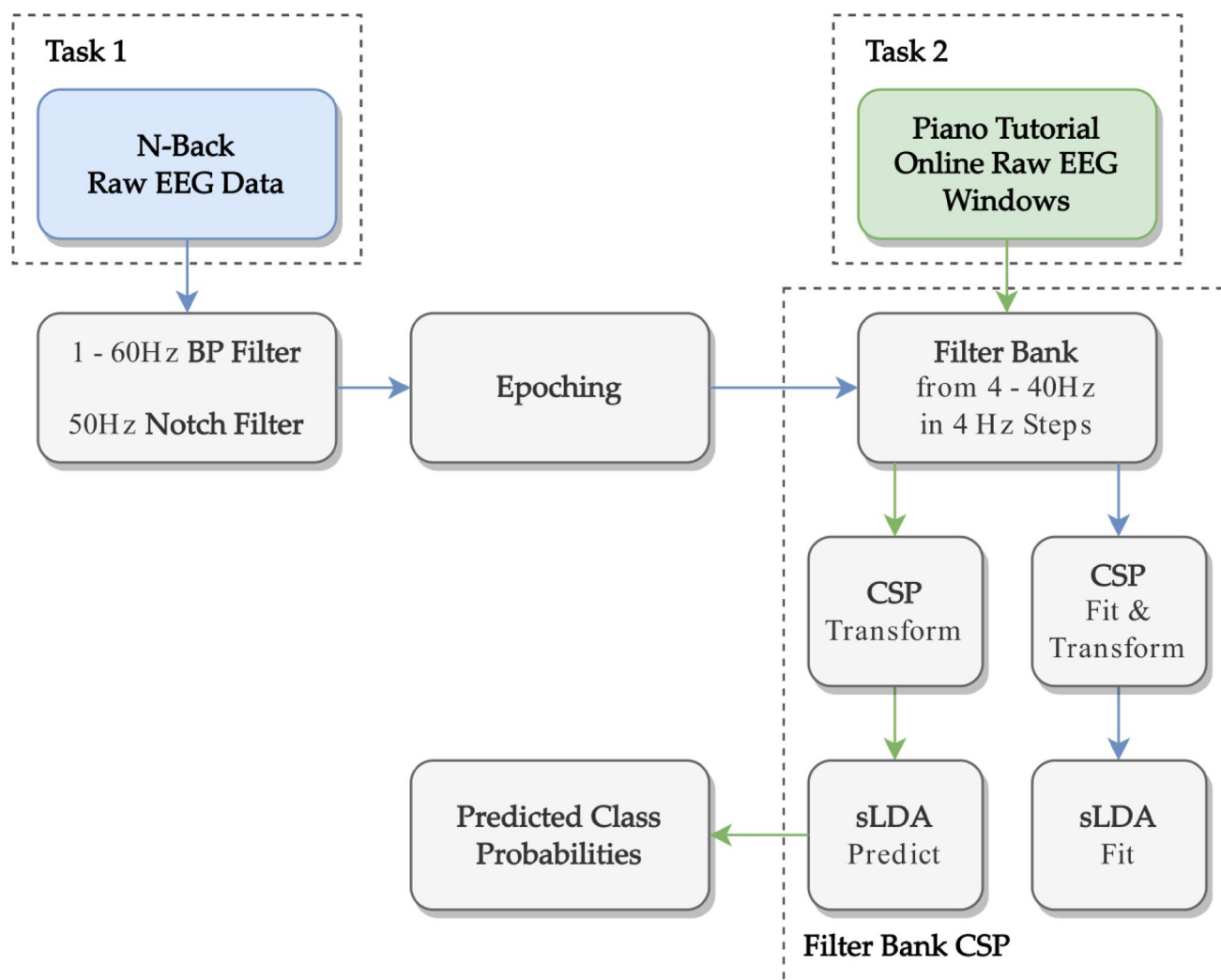


Fig. 7 Signal processing chain

1-back data was not involved in the training of the FBCSP and only used for later analysis. Since one n-back run (without instructions) lasted 50 s, 23 overlapping windows were recorded. The number of windows was chosen, so that all of them are within the 50 s run time frame. Since a part of the 24th window would already be outside of the time frame, this data was scrapped. Since there were 4 runs per condition, that made for a total of 92 possible epochs per condition. Epochs with a maximum peak-to-peak signal amplitude of more than 120 μV in any channel were automatically rejected. Values above 120 μV mostly do not correspond to the EEG and stem often from muscle artifacts or eye blinking.

2.5.3 Filter Bank Common Spatial Pattern (FBCSP) algorithm

As illustrated in Fig. 2, the n-back task (task 1) and the piano tutorial (task 2) were interconnected by the FBCSP algorithm which was implemented to achieve an online MWL detection. The Filter Bank (FB) is the first step of the FBCSP algorithm and was used to divide both the n-back and the piano tutorial data into several frequency bands, correlated to MWL.

Borghini et al. (2014) and Babiloni (2019) claimed that changes in the theta band (4–8 Hz) and alpha band (8–12 Hz) are associated with high or low MWL. In more detail, a rise in BP at frontal cortical areas in the theta band with a simultaneous decrease in BP at parietal cortical areas in the alpha band are associated with an increase in MWL.

Fluctuations in the beta band (12–30 Hz) and lower gamma band (30–60 Hz) have also been correlated to changes in

MWL (Brookings et al. 1996; Howard et al. 2003; Dimitrakopoulos et al. 2017; Roy et al. 2016; Hamann and Carstengerdes 2023).

Following the work of Arvaneh et al. (2015), we selected nine broad frequency bands with a range from 4 Hz to 40 Hz (4–8 Hz, 8–12 Hz, 12–16 Hz, 16–20 Hz, 20–24 Hz, 24–28 Hz, 28–32 Hz, 32–36 Hz, 36–40 Hz). After completing the n-back task, each epoched sub-band underwent fitting to a Common Spatial Pattern (CSP) class for the estimation of spatial filters used in feature extraction. The number of filters employed to decompose the EEG signal was set to 6 for all experiments, and features underwent log transformation for standardization. Subsequently, the n-back data was transformed, resulting in $nf_bands * nfilters$ features per epoch. With 9 frequency bands and 6 spatial filters, this led to 54 features per epoch. The CSP-filtered features for all sub-bands and epochs were then fit to the final step of the FBCSP algorithm, the sLDA classifier. As only data from the 0 - and 2 - back runs were utilized, the sLDA classifier was trained as a binary classifier to distinguish between low (0-back) and high (2-back) Mental Workload (MWL). As participants began the piano tutorial, a new EEG window was sent through the trained FBCSP algorithm every 2.5 s. However, the algorithm was solely employed for classifying incoming online data. Features were extracted using the CSP algorithm and then predicted by the sLDA classifier. The prediction encompassed the class label (low or high MWL) along with the estimated probability of a sample belonging to that class. In the adaptive participant group, this probability was utilized to determine, at the end of each level, which song would be played in the subsequent level.

3 Data analysis

After the experiment, the recorded and pre-processed EEG data, as well as all the recorded data from the questionnaires were used for further analysis. This included the score of the 10-fold cross-validation, the performance during the comparison songs before and after the piano tutorial, the average classified MWL per level, and the scores for the NASA-TLX and SUS questionnaires.

3.1 Piano key accuracy

To evaluate the participant's performance during both comparison song sections, their accuracy in terms of correctly pressed piano keys was evaluated. Therefore, a script was written, which considered a key correctly hit, if (a) the key

was pressed not earlier than 200 ms before the holographic key touched the green line and (b) the key was pressed not later than 500 ms after the holographic key touched the green line. Those limits were chosen after different pre-tests of the piano tutorial were conducted where the player was mostly hitting the correct notes and therefore achieved a score of higher than 95%.

3.2 Mental Workload and Mental Fatigue Band power analysis

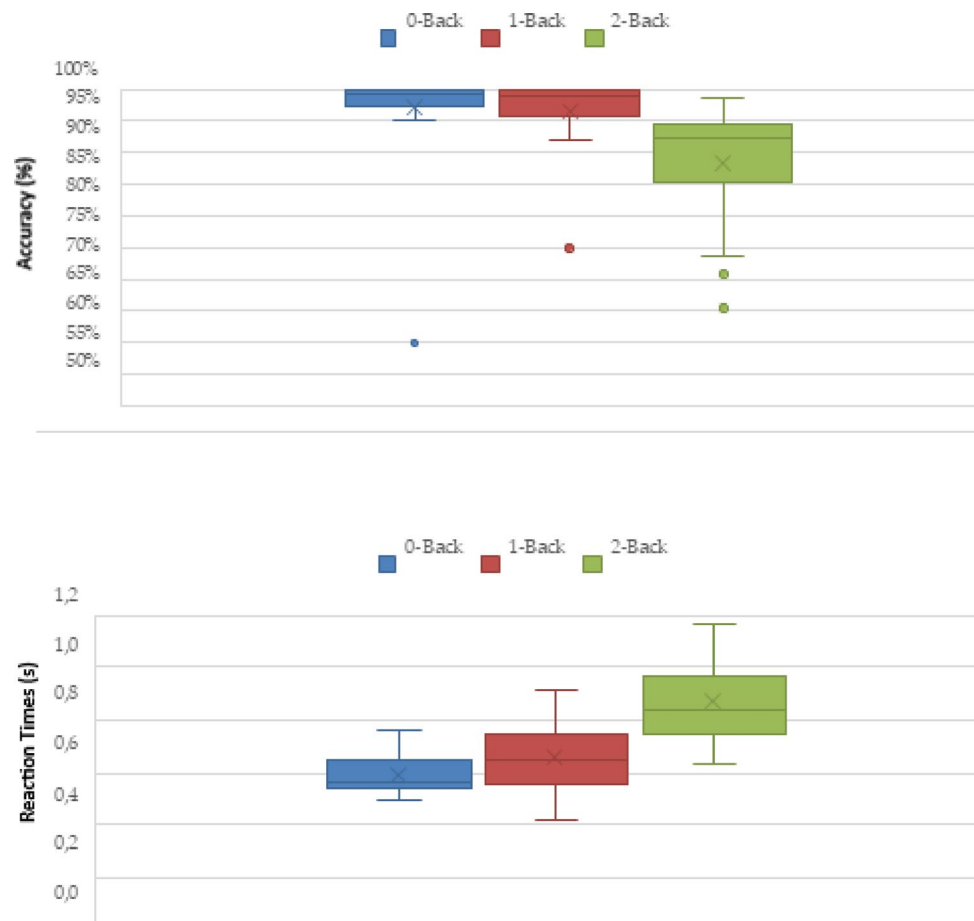
Whereas in the online tutorial only MWL was classified, in the offline analysis MWL and MF was analysed based on the change of relative BP in different frequency bands and cortical areas during the piano tutorial. For example, Klimesch et al. (1999) associated an increase in the BP frequency bands below 12 Hz. Specifically in the Theta band (4–8 Hz) an increase reflects elevated cognitive control effort and reduced alertness. Also increases in BP in the alpha band (8–13 Hz) are highly relevant indicating reduced cortical activity and increased inhibition of task-irrelevant regions with reduced processing capacity. On the other hand, a simultaneous BP decrease in the frequency band higher than 12 Hz during MF has been observed. For example, reduced alertness and lower active cognitive engagement can be found in the beta band (13–30 Hz). Babiloni (2019) suggested, that an increase in MWL is associated with a band power decrease in the alpha frequency band (8–12 Hz) at parietal cortical areas and an increase in the theta frequency band (4–8 Hz) at frontal cortical areas. Following that the frequency bands reflecting MF and MWL have been chosen accordingly. The slope of the linear regression of the relative BP was calculated for every participant. After the recording of the raw data, every level was pre-processed (see 2.5 *Signal Processing*). Subsequently, a time window of 5 s length was defined, which was moved across the pre-processed EEG of each level in steps of 200 ms and used to calculate the relative BP during its span. Thus, for each window the power spectral density (PSD) was calculated using the multitaper method. Multitaper is a widely used technique to estimate the spectral representation of a system based on finite observations (Babadi and Brown 2014). The PSD was normalized by the sampling rate and the length of the signal. To generate the normal BP, each window was integrated using the Simpson's rule over the frequencies of interest. These frequencies included the theta band (4–8 Hz), the alpha band (8–12 Hz), as well as the frequency region of 1 to 12 Hz and 12 to 40 Hz. To obtain the relative BP, each BP of the frequency regions of interest was divided by the BP of the whole frequency spectrum, which was limited from 1 to 40 Hz due to the pre-processing step. Subsequently, those

values were fit to a linear regression model and the slope of the resulting line was calculated as seen in Eq. (1).

$$k_{level} := \frac{\Delta y_{lr}}{\Delta x_{step}} \quad (1)$$

k_{level} is the slope of the linear regression per level, Δy_{lr} is the difference between two adjacent points of the linear regression, while Δx_{step} is the window step length of 200 ms. This resulted in 32 values, one for each electrode. To compare the results over certain brain regions, the slopes over the electrodes were averaged. For the theta band, the electrodes of the frontal cortex (Fp1, Fz, F3, F7, FT9, FC5, FC1, FC6, FC2, F4, F8, Fp2), for the alpha band, the electrodes of the parietal cortex (CP5, CP1, Pz, P3, P7, P4, P8, CP6, CP2) and for the 1 to 12 Hz and 12 to 40 Hz frequency bands all electrodes were used for averaging. The positions for the alpha and beta band were chosen based on findings by Babiloni (2019). For further comparison, the computed average slope values were averaged among the participants and based on their group membership.

Fig. 8 Boxplots of the n-back performance accuracies (=correctly classified target and non-target letters) and reaction times (=individual reactions to the letters per condition)



4 Results

4.1 Cross-validation of the FBCSP algorithm

Regarding the used n-back task, the reaction times and performance accuracies for the different conditions of the n-back task were analysed to determine their correlation with the mental workload (MWL) they induced (Zhang et al. 2018). Previous studies on the n-back task have consistently shown that as task difficulty increases, reaction times (RT) tend to increase, while performance accuracy decreases (Babadi and Brown 2014). The condition with the lowest MWL would be expected to yield the highest performance accuracy and the shortest reaction times.

The highest performance, with a mean of 97.16%, was the 0-back task (Fig. 8). Except for one outlier of 60%, all values were ranging from 95 to 100%. The 1-back task yielded the second highest average performance with 96.71% and the lowest performance accuracies were gathered from the 2-back task with an average of 88.38%. The RT is the time after participants saw a letter until they pressed a piano key for the first time. The times were sorted according to their n-back condition and averaged over all letters of all four runs per condition. The 0-back task was performed

in average with the shortest reaction time of 0.59 s, ranging from 0.49 to 0.76 s. The second lowest average reaction time was yielded during the 1 -back task with 0.65 s and a range from 0.42 to 0.91 s. The 2 -back task had the highest average reaction time of 0.87 s ranging from 0.63 to 1.16 s. For the performance accuracies, a significant difference was found between the n-back conditions [$\chi^2=30.1$, $df=2$, $p<0.001$, $\epsilon^2=0.463$]. In the pair-wise comparison, the difference between 0 -back and 2 -back [$W=-6.96$, $p<0.001$] and 1-back and 2-back [$W=-6.96$, $p<0.001$] were significant. 0 -back versus 1-back yielded no statistical significance [$W=-2.3$, $p=0.235$]. For the reaction times, the Kruskal-Wallis ANOVA revealed again a statistical significance [$\chi^2=31.2$, $df=2$, $p<0.001$, $\epsilon^2=0.479$]. In the pair-wise comparison, the difference between 0-back and 2-back [$W=7.31$, $p<0.001$] and 1-back and 2 -back [$W=5.63$, $p<0.001$] were significant, while 0-back and 1-back yielded no statistical significance [$W=2.86$, $p=0.107$].

These results validate the use of the 0-back and 2-back tasks as low and high MWL conditions, respectively (Fig. 8).

To test the performance of the FBCSP algorithm, a 10-fold cross-validation was conducted after training with the n-back participant data. The classifier algorithm was trained and tested to discriminate between epochs of the 0- and 2-back task. The mean cross-validation score was 71.57% with results ranging from 45.36 to 93.24%.

4.2 Performance before and after the Piano Tutorial

Both comparison songs, before and after the piano tutorial, were evaluated in terms of performance accuracy. For the adaptive group, a mean of 70.12% pre- and 76.68% post-tutorial was measured, which was an improvement of 6.56%. Performance accuracies ranged from 60 to 82.61% and from 63.48 to 89.57% for pre- and post-tutorial, respectively. The non-adaptive group had a mean of 64.53% pre- and 77.65% post-tutorial for the performance accuracies, which was the highest improvement of 13.12%. Performance accuracies ranged from 44.35 to 78.26% and from 66.09 to 89.57% for pre- and post-tutorial, respectively. The performance changes between the comparison songs were evaluated for statistical significance for each group individually. For the adaptive group, no statistical significance was found [$t=-1.84$, $df=10$, $p=0.096$], while for the non-adaptive group, a statistical significance was found [$t=-3.78$, $df=10$, $p=0.004$].

4.3 Performance during the piano tutorial

Each of the 10 levels of the piano tutorial were evaluated in terms of performance accuracy.

Figure 9a, b illustrating the performance of participants in terms of correctly hit piano keys during the tutorial for the adaptive and non-adaptive group respectively (Figs. 10, 11, 12, 13 and 14).

Each level of the piano tutorial was evaluated for performance accuracy in the same way. Figure 15 illustrates the performance for all participants and per level for the adaptive (9a) and non-adaptive group (9b). The progression of the performance accuracy, averaged over the participants for both groups is visualised in Fig. 10. The performance accuracy was fit for linear regression. The regression equations are listed in Eqs. 2 and 3, for the adaptive and non-adaptive group, respectively. The constant (y-distance) between the two regression lines was 5.75% and not statistically significant [$p=0.291$]. The coefficient (slope) difference of 1.55% was also not statistically significant [$p=0.086$].

4.4 Mental Workload classification

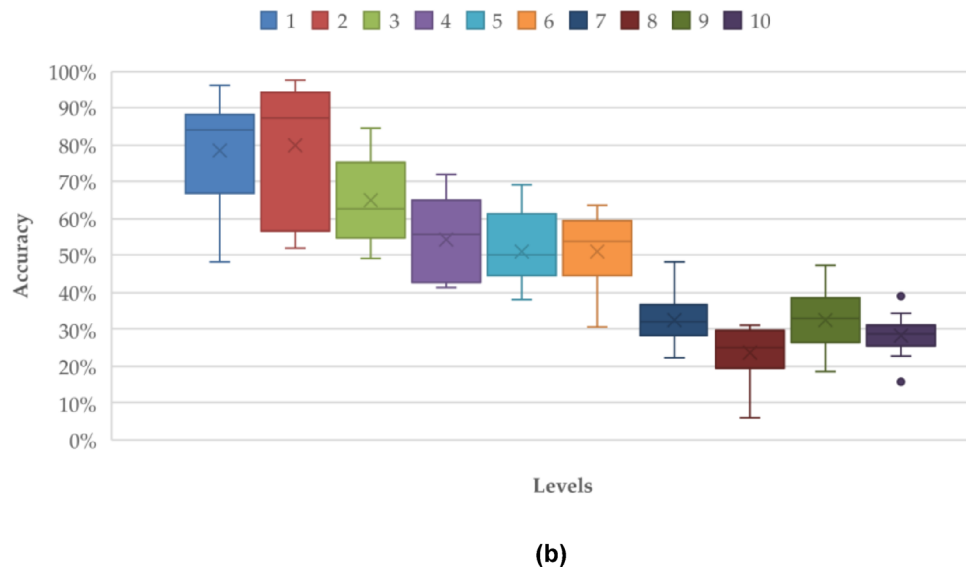
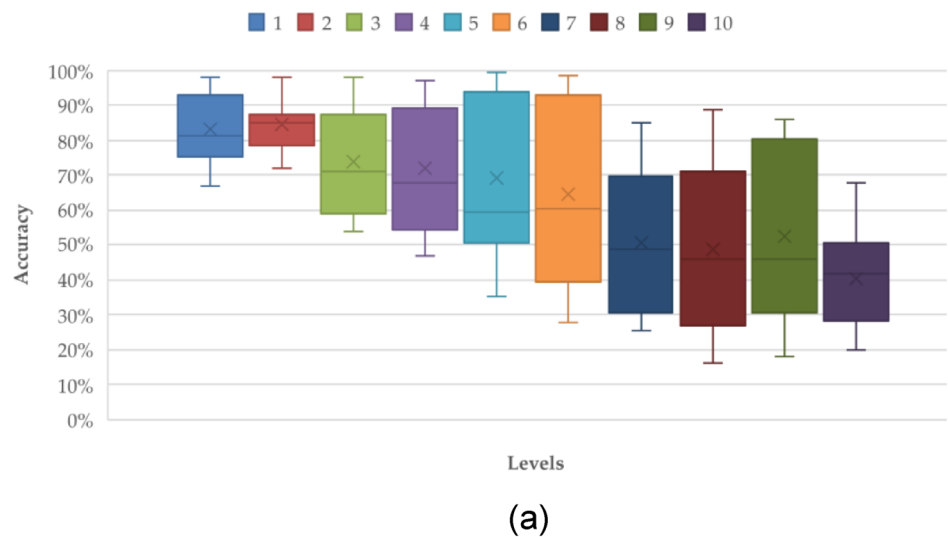
The FBCSP algorithm, trained on n-back data, performed online predictions of MWL windows at 2.5-second intervals across all levels of the piano tutorial. The results were expressed as values ranging from 0 to 100%, where 0% indicated a definite association with the low MWL class (as measured in the 0-back task), and 100% indicated a definite association with the high MWL class (as measured in the 2-back task). In Fig. 11, the average MWL per level is illustrated, along with their linear regression fits, for both the adaptive and non-adaptive group.

A linear fit was conducted for both groups. The average MWL of the adaptive group consistently remained lower than that of the non-adaptive group across all levels, with percentages ranging from 43.47 to 57.11% and 63.44% to 75.75% for the adaptive and non-adaptive groups, respectively.

The average classified MWL per level was fit to a linear regression model and then compared between the two groups. The constant difference was -22.9% and statistically significant [$p=0.000$]. The difference in coefficient was 0.351% and not statistically significant [$p=0.607$].

In Fig. 12, the level progress of the adaptive group is shown. Each column corresponds to one participant of the adaptive group. After each level, the results of the online classification (either low or high MWL) decided, whether participants played the next harder song or replayed the same song. After three levels with a high MWL, participants continued with the next harder song, not taking the result of the classifier pipeline into account. Each participant progressed from level 1 to 10 throughout the piano tutorial, but unlike the non-adaptive group, the level number was not necessarily identical to the song number. A rising song number meant a higher song complexity and difficulty. Each column

Fig. 9 **a** Boxplots illustrating the performance of participants in terms of correctly hit piano keys during the piano tutorial for the adaptive group. Each boxplot corresponds to the performance during one level (blue - level 1, red - level 2, ...) **b** Boxplots illustrating the performance of participants in terms of correctly hit piano keys during the piano tutorial for the non-adaptive group



in the diagram represents one participant. After three levels of high average MWL, the next harder song was chosen for the next level.

4.5 Relative band power changes - mental fatigue detection

To further assess the Mental Fatigue (MF), we examined the average change in relative Band Power (BP) at each level. This change was identified by fitting a linear regression to the trajectory of the relative band power during the different levels of the piano tutorial, as explained in the Mental Workload and Mental Fatigue Band Power Analysis. In Fig. 13, the alterations in relative band power for the 1 to 12 Hz band are depicted, along with the linear regression

of these changes across all electrodes throughout the levels. In Fig. 10B the alterations in relative band power for the 12 to 40 Hz band are depicted, along with the linear regression of these changes across all electrodes throughout the levels. In the adaptive group, a positive average relative BP change for the 1–12 Hz band was observed at levels 4, 7, 8, and 10. For the linear regression, both the adaptive and non-adaptive group showed a positive linear trend, with the slope being steeper for the adaptive group with 0.00314 1/min compared to 0.00068 1/min of the non-adaptive group. Comparing the adaptive against the non-adaptive regression model, both the difference between constants (-0.0348 1/min, [$p=0.074$]) and slopes (0.00246 1/min, [$p=0.414$]) were not statistically significant.

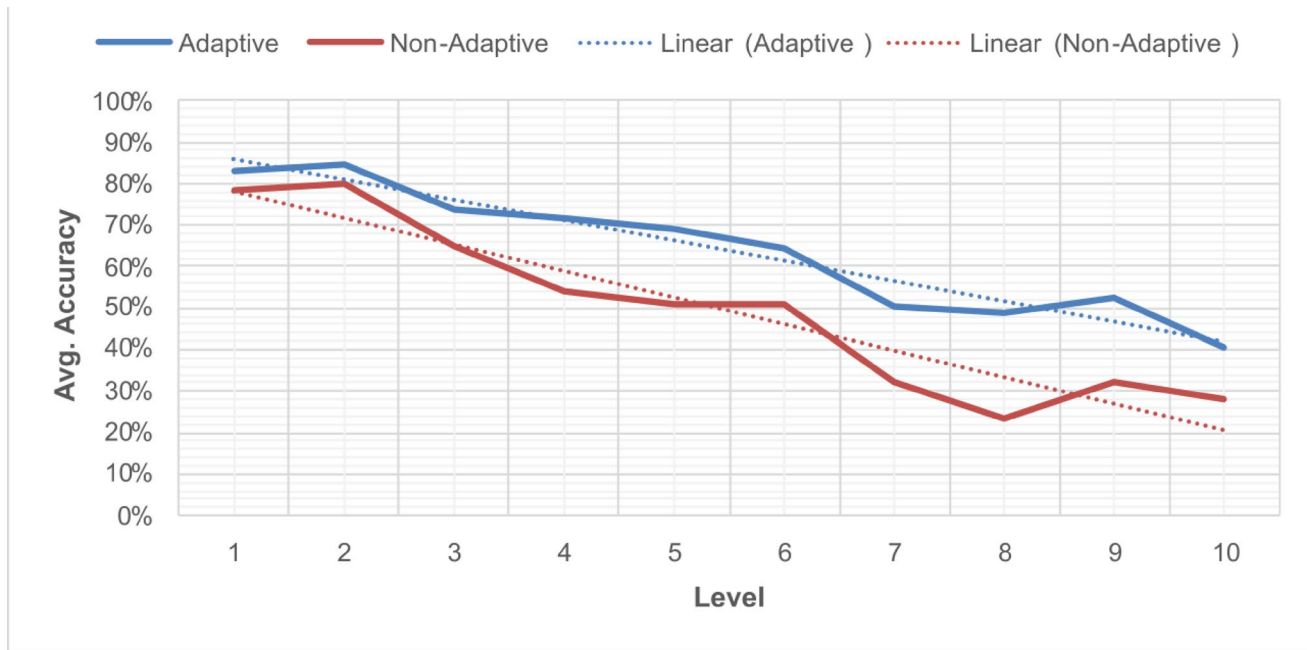


Fig. 10 Average performance accuracy for both groups at all 10 levels

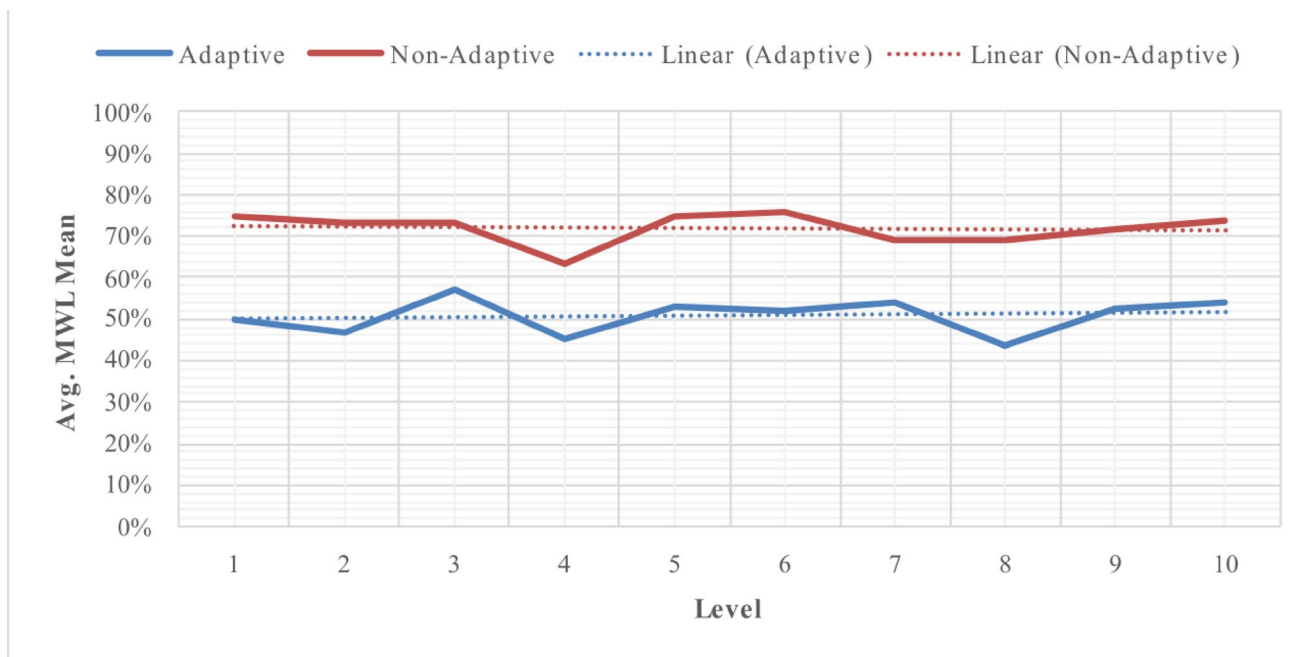


Fig. 11 Average mean MWL for both groups at all 10 levels

The results of the 10×2 repeated measures ANOVA (RM-ANOVA) including the within factors “song level” (Level 1–10) and “frequency bands” (1–12 Hz/12–40 Hz) and the between factor “group” (adaptive/non-adaptive) showed no significant differences.

4.6 Relative band power changes - Mental Workload Detection

Figure 11 illustrates the relative band power changes for the theta (4–8 Hz) and alpha (8–12 Hz) frequency bands as well as the linear regression of the changes throughout the levels for both groups (11 A: adaptive group; 11B: non-adaptive group). In the theta relative BP calculation, only electrodes

Fig. 12 The level progress of participants during the piano tutorial

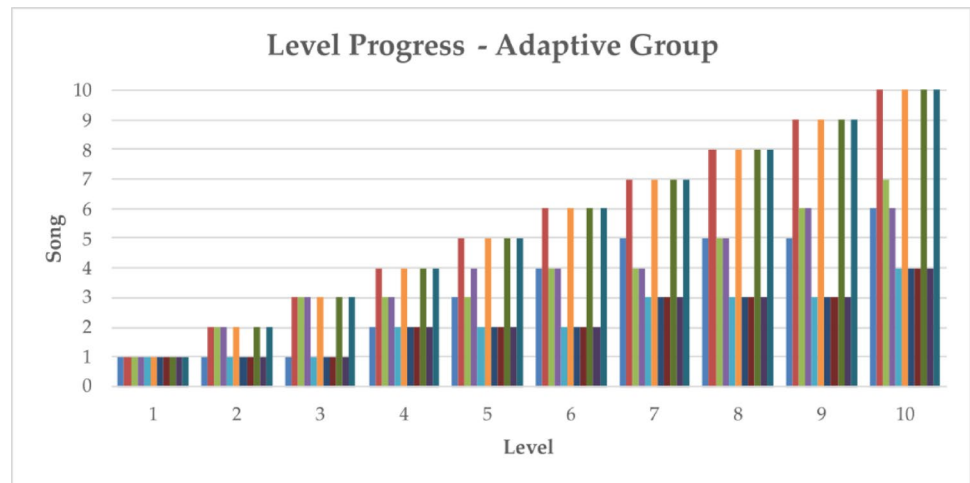
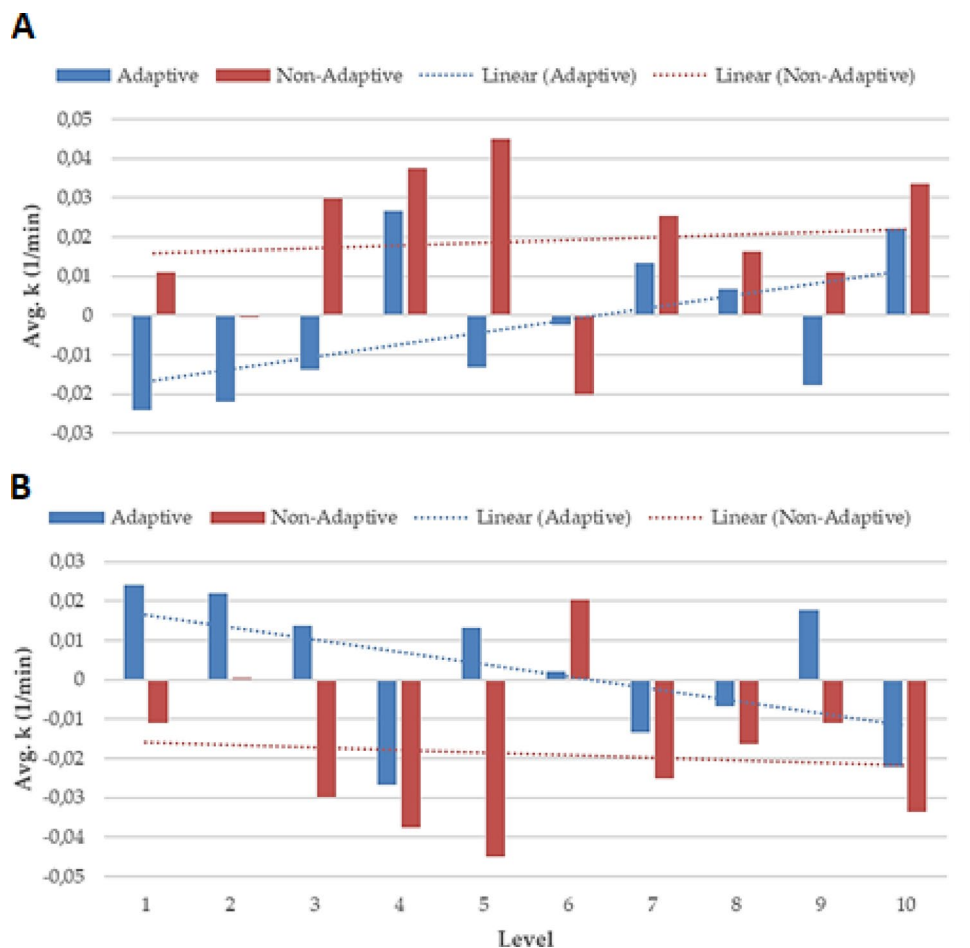


Fig. 13 Average change of relative band power in MF per level for both groups. **A** 1 to 12 Hz and **B** 12 to 40 Hz



from the frontal cortical areas and for the alpha band, electrodes from the parietal cortical areas were used.

A simultaneous relative BP increase in the theta frequency band at frontal cortical areas and relative BP decrease in the alpha band at parietal cortical areas was detected for the adaptive group at levels 6, 7 and 9 and for the non-adaptive group at levels 4, 8 and 9. In the adaptive group, the alpha

band showed a negative trend ($-0.00237 \frac{1}{min}$) and the

theta band a positive trend ($0.000683 \frac{1}{min}$), with the alpha band slope being steeper and having a higher constant. In the non-adaptive group, the alpha band revealed a negative

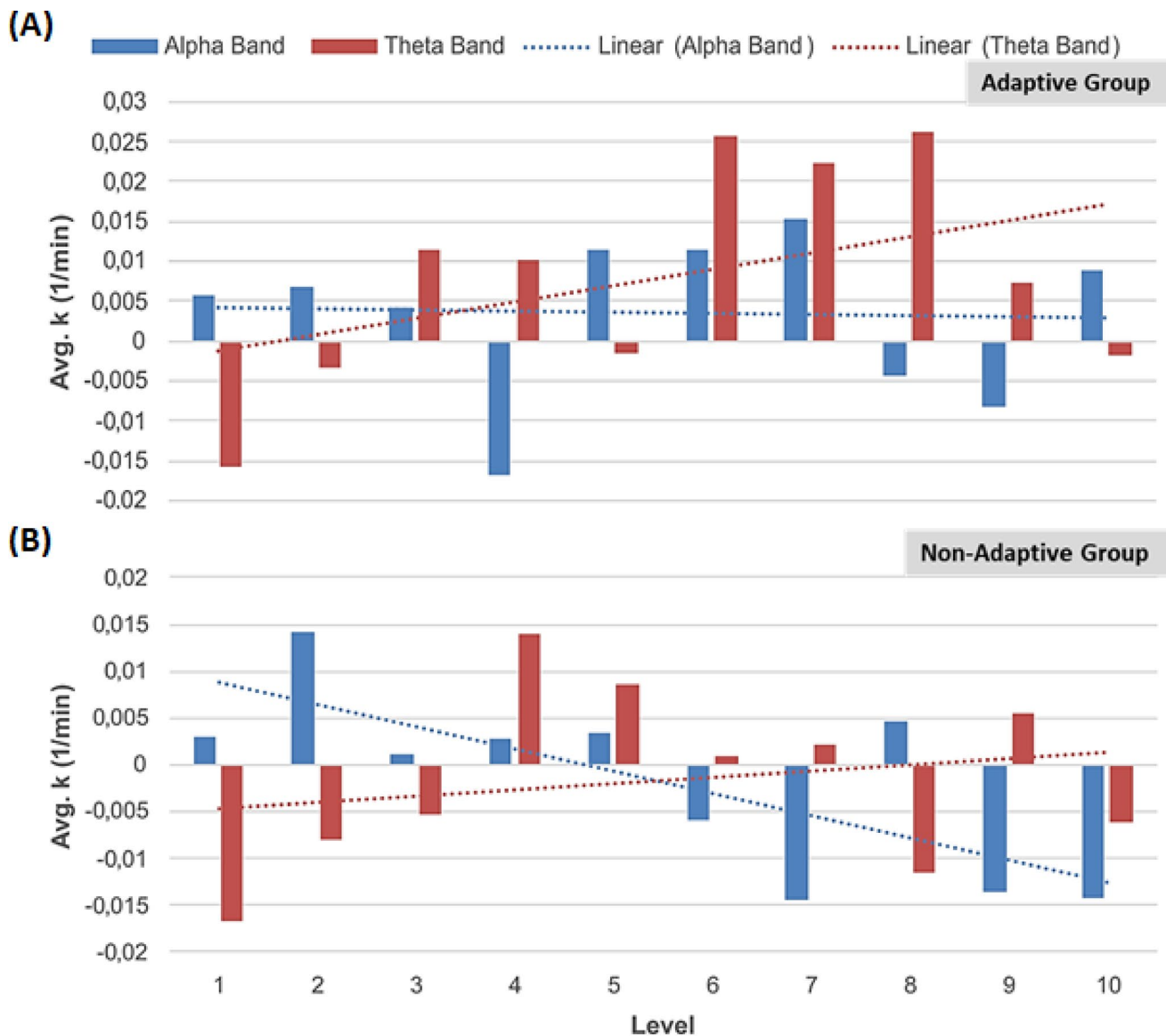


Fig. 14 Average change of relative band power in MWL per level comparing the alpha (8–12 Hz) and the theta (4–8 Hz) band for the adaptive (A) and non-adaptive group (B)

trend ($-0.00016 \frac{1}{min}$) and the theta band a positive trend ($-0.00205 \frac{1}{min}$).

The results of the 10×2 repeated measures ANOVA (RM-ANOVA) including the within factors “song level” (Level 1–10) and “frequency bands” (alpha/theta) and the between factor “group” (adaptive/non-adaptive) showed no significant differences.

4.7 Results of questionnaires

(a) NASA task load index

The NASA-TLX is a questionnaire to measure the subjective MWL of participants. The overall mean of the adaptive group was slightly higher with 68.33 than the non-adaptive group with 67.88. The performed paired samples t-test revealed no statistical difference for the means of the NASA-TLX score between the adaptive and non-adaptive group [$t=0.122$, $df=20$, $p=0.904$]. Taking a closer look at the different subscales, the Onaway ANOVA revealed a significant difference between the groups only for the factor “mental demand” [$F=7.9$ (1,17), $p=0.012$]. Figure 15 illustrates the detailed results of the different subscales for each group separately.

(b) System usability score

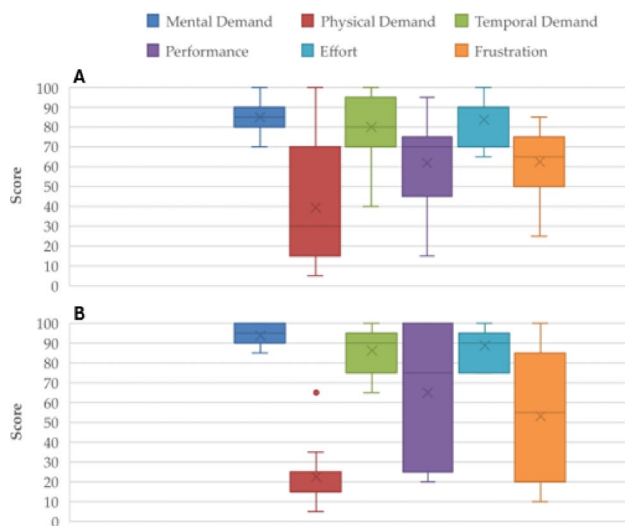


Fig. 15 Boxplots of the individual NASA-TLX scales. **A** Adaptive group; **B** Non-adaptive group

To get feedback on the usability and functionality of the AR piano tutorial, participants filled out the SUS. No significant differences between the groups were observed (adaptive group: mean=74.8, SD=13.4; non-adaptive group: mean=71.2, SD=15.6). An average score of 73.07 was achieved, indicating an overall good usability of the tutorial.

5 Discussion

The objective of this study was first to combine BCI technology with an AR application in online mode and second to identify varying levels of Mental Workload (MWL) to enhance the performance of individuals undergoing a piano tutorial. To substantiate the findings of the classification, a relative Band Power (BP) analysis was carried out to identify shifts in MWL and Mental Fatigue (MF). Whereas MF was only used for data analyses, the detected MWL was used for online adaptation of the system. Additionally, participants' behavioural data were assessed based on their performance during the piano tutorial and comparison songs. Additionally, subjective measures were implemented through two questionnaires (NASA-TLX and SUS). We were able to meet our first aim, namely to fuse a pBCI with AR technology in a neuro-adaptive system, but the overall efficiency of the neuroadaptation was unfortunately not reached. The efficacy of the FBCSP algorithm was assessed through a 10-fold cross-validation using two classes from the n-back data: 0-back and 2-back. The mean cross-validation score attains 71.57%, surpassing chance levels. This confirms the classifier's consistent ability to distinguish between the 0-back and 2-back classes.

5.1 MWL and MF band power analysis

From a theoretical perspective, the EEG results align with established neurocognitive models of mental workload and fatigue. In the non-adaptive group, the progressive increase in relative theta band power across levels, together with stable or slightly elevated alpha power, is consistent with sustained working-memory engagement and accumulating mental fatigue, as frontal-midline theta has been repeatedly linked to increased cognitive control demands and effortful processing (Gevins and Smith 2003; Cavanagh and Frank 2014), while reduced alpha suppression over time is often interpreted as diminished cortical efficiency under prolonged load (Klimesch 1999; Wascher et al. 2014). In contrast, the adaptive group showed a comparatively flatter theta trajectory and a decreasing alpha trend, suggesting more efficient allocation of cognitive resources and reduced fatigue accumulation across the tutorial. Importantly, these differences manifest primarily as divergent temporal dynamics rather than large mean-level separations, which is theoretically consistent with adaptive workload regulation models that predict stabilization of cognitive effort over time rather than absolute reductions in instantaneous demand (Hancock and Warm 1989; Hockey 1997). This dynamic interpretation also explains why the RM-ANOVA on band power did not yield statistically significant group effects, despite converging subjective (NASA-TLX Mental Demand) and electrophysiological indications that the non-adaptive group experienced greater sustained cognitive load.

The expected accelerated increase in Mental Workload (MWL) in the non-adaptive group did not occur, revealing only a slightly diminishing pattern. Linking low performance to high MWL, when considering performance accuracies across levels, did not demonstrate a corresponding elevation in MWL. Possible explanations for the consistently classified MWL include participants generating EEG artefacts while playing the piano. Despite instructions to play calmly, maintaining ideal EEG conditions proved challenging. Artefacts were not filtered out during online classification trials, potentially causing the classifier to capture noise rather than accurately detecting MWL, which is a serious lack of the study. Additionally, the substantial difference between the training task and the piano tutorial may have impacted the classifier's ability to detect MWL changes as hypothesized. Effective cross-task classification requires the training task to induce a similar type of MWL as the target task, potentially contributing to unexpected results. According to Klimesch et al. (1999) an increase in the BP frequency bands below 12 Hz and a simultaneous BP decrease in the frequency band higher than 12 Hz is associated with MF. The relative BP analysis for the 1 to 12 Hz frequency band revealed that the adaptive group showed an

increase in relative BP in the 1–12 Hz band in levels 4, 7, 8 and 10, indicating increased MF especially in the latter levels of the tutorial. For the non-adaptive group, signs of MF according to the aforementioned definition were detected in all levels except for level 2 and 6. This inferred that the adaptive group experienced less MF over the tutorial than the non-adaptive group, suggesting that the adaptive difficulty helped reduce MF (Fig. 10). Also, the increased MF in the later stages of the piano tutorial correlated with the increased time-on-task participants were exposed to. Because of the band pass filtering of 1 to 40 Hz, the results of the 12–40 Hz band were reciprocal to the results from the 1–12 Hz group. Babiloni (2019) suggested that an increase in MWL is associated with a band power decrease in the alpha frequency band (8–12 Hz) at parietal cortical areas and an increase in the theta frequency band (4–8 Hz) at frontal cortical areas. The linear regressions for the theta and alpha relative BP changes suggest that for the adaptive group, there was a strong trend towards negative alpha relative BP change and a moderately steep rise of the change of the theta relative BP (Fig. 11). This could indicate that in the latter level MWL tends to be stronger in the participants, which is coincidental with the performance measures during the tutorial. For the non-adaptive group, there was a strong rise of theta relative BP change. The linear regression of the alpha relative BP change showed a slight negative trend. Again, this could imply that there is a rise in MWL, the more the participants progressed through the levels, especially considering the theta levels. So, in conclusion, in both groups a rise in MWL was detected towards the latter levels. The adaptive group showed a steeper fall in the alpha band, while the non-adaptive group revealed a more significant increase in the theta band.

5.2 Behavioural results

The behavioural analysis indicated that the n-back task induced varying levels of mental workload (MWL) across tasks, with the 0-back task associated with low MWL and the 2-back task associated with high MWL. The classifier was trained sufficiently well to distinguish between epochs of the 0-back and 2-back tasks. Contrary to expectations, no significant learning effect was observed in participants' performance before and after the tutorial. Only the non-adaptive group demonstrated a significant improvement in performance accuracy during the comparison songs, although both groups showed some overall improvement.

By comparing the performance before and after the tutorial for the same song, both groups exhibited improvement on average, indicating the tutorial's efficacy in enhancing piano skills. Contrary to the hypothesis, the non-adaptive group demonstrated a larger improvement (13.12%)

compared to the adaptive group (6.56%), despite the latter's level adjustments based on measured MWL. Post-tutorial performance levels were similar between the two groups (77.65% and 76.68%). Significance in performance change was confirmed only for the non-adaptive group, challenging the anticipated advantage of the tutorial for the adaptive group. This suggests that participants, especially those with consistently high MWL, have played mostly easier songs, limiting their skill improvement. The better performance of the non-adaptive group might also be due to the more rudimentary system model, repeating songs just based on high MWL.

As depicted in Fig. 8, both groups exhibited, on average, a negative linear trend across the levels. However, neither the disparity in the constant nor the slope was statistically significant. This suggests that the performance difference between the two groups was not significant, and consequently, the designed levels did not attain the hypothesized distinction between the two groups. Figure 9 depicts the average MWL per level for the adaptive and non-adaptive groups, as well as the means and linear regression of the average MWL across both groups. The regression analysis indicates a significant increase in MWL from the adaptive to the non-adaptive group in terms of the coefficient, though the comparison of slopes shows no significance. This implies that the non-adaptive group experienced higher induced MWL across levels. The observed higher MWL in the first level of the non-adaptive group, where both groups played the same song could be explained by experience. We assume that some people had already experience with this song and were unluckily assigned to the same group. We have asked the participants before regarding their experiences with AR but neglected to ask about their musical abilities, specifically if they are able to play another instrument. We only took care that none of the participants had any prior knowledge of playing piano, that they were all novices. So, if the non-adaptive group contains more overall music novices, it will explain the gap. Even with proper random assignment, such imbalances happen, especially with modest N. Beside the unequal prior music experience also cognitive or affective state differences at arrival (e.g. stress, sleep debt, caffeine, anxiety) might play a role. But also, possible trait differences like working memory capacity, neuroticism, "tech anxiety," or motivation should not be neglected. All factors could have contributed to the initial group differences, specifically in such a small sample size.

Performance accuracy throughout the piano tutorial levels revealed a steady decline for both groups. However, no significant difference was found between the adaptive and non-adaptive groups, suggesting that the adaptive group's tutorial should have been simplified or modified to sustain consistent performance. Results from the NASA-TLX

questionnaire indicated that the piano tutorial elicited moderate to high MWL across both groups, with no significant difference in overall NASA-TLX scores between the groups. Except the subscale “mental demand” significantly differed between the groups, showing higher scores for the non-adaptive group. This finding aligns directly with the objective measurement of significantly higher classified MWL in the non-adaptive group. A possible explanation could be that the task itself required more conscious control since without adaptation participants likely had to monitor errors more closely, consciously regulate their pacing or compensate for difficulty mismatches. All together increases executive control and working-memory engagement which is exactly what the “Mental demand” subscale captures. Another reason for the higher experienced mental workload in the non-adaptive group could be a poorer task-skill matching provided by the tutorial. Since adaptive systems aim to keep task difficulty aligned with user ability. A higher Mental Demand suggests that for the non-adaptive group the task was too difficult for some participants or fluctuated around their capability without adjustment and therefore forcing them to “work harder” mentally to keep up. Concluding, the higher mental demand ratings in the non-adaptive group suggest that participants perceived the task as more cognitively taxing, likely reflecting increased executive control, sustained attention, or compensatory effort in the absence of task adaptation. This indicates reduced cognitive efficiency rather than necessarily poorer performance. The SUS questionnaire indicated that the Piano Tutorial application was rated above average in usability in both groups.

6 Challenges and future work

For the first time we combined an online pBCI system with an AR application which delivers on the one side interesting opportunities but on the other side opens novel challenges and limitations. Starting with the small sample size of the study which is without any doubts a limitation. Since only 11 participants per group were analysed it cannot be finally claimed that the neuro-adaptive approach failed at all. Future work on a huge sample size is absolutely necessary. Nonetheless our findings might be interesting for researchers in the field and could contribute to the development of future neuroadaptive systems. Another drawback was the pre-processing for classification. For the online classification, no artefact correction, except for band pass filtering, was implemented, which led to bigger artefacts being classified. Multiple online artefact correction tools exist already, with Independent Component Analysis (ICA) being one of the most used ones, for example by Mayeli et al. (2016) and Bhardwaj et al. (2015). ICA involves

multiple measurements of biomedical signals, where statistically independent sources are extracted and separated (James and Hesse 2005). However, this process uses a lot of computational resources and was therefore excluded in this experiment. In future studies, increased computational power could solve this problem. Also, the online classification could be adapted in the tutorial in a more fluid way leading to immediate reactions according to the mental state of the user.

During the measurements, a few participants had to put off their glasses because they did not fit into the MR goggles, causing worse visibility. For future studies, larger MR headsets or specific adapters could be used to solve this problem. Another issue included the visibility of the holographic piano keys. They were color-coded in a brighter and darker green. Two participants with colour blindness could not separate them from each other. This could have been easily avoided by choosing more distinctive colours. Additionally, many participants reported that the frontal electrodes (FP1 and FP2) caused pain, because the Meta Quest 2 pressed directly onto them. However, since the frontal electrodes were important, especially for the theta band analysis, this could not have been avoided with this setup. Differently shaped MR goggles could bypass this problem.

Despite the described limitations we can state that the fusion of an online BCI and AR technology was successful and a promising approach on the way to develop more engaging and adaptive learning environments. Reporting also non-significant results improve reproducibility by identifying model-based or context-dependent effects. Furthermore, the idea to integrate the actual mental state of the users allows us to provide feedback that is tailored to the individual, by selecting the proper complexity of songs.

7 Conclusion

Overall, our approach showed promising results for future neuroadaptive piano tutorials and tutorials in general. Even though a clear advantage for the adaptive group was not observed, this study could pave the way for more personalized learning environments or online tutorials, introducing innovative approaches for technology to deliver information and assist individuals in their daily lives. For future studies we would suggest an improved design for the adaptive group, for example by introducing a tutorial that changes the difficulty while participants are still playing and not only discreetly per level. Also, a sophisticated online artefact correction would help filter out noise and increase classification performance. Finally, an increased number of participants would improve the validity of the experiment.

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Declarations

Competing interests The authors declare no competing interests.

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